

Madhav Trivedi

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SUMMARY

Software Engineer with 2+ years of full-time experience building production applications using C#, .NET, Angular and AWS. Experienced in developing REST APIs and asynchronous services, optimizing SQL performance and improving application reliability. Built and load-tested backend projects focused on transaction integrity, real-time concurrency and automated testing using PostgreSQL, RabbitMQ and Redis.

EXPERIENCE

Zeus Learning

Software Engineer

Jan 2024 – Present

Mumbai, India

- Shipped 2 .NET 8 RabbitMQ export services with resumable checkpoints, retries, dead-letter queues and alerts.
- Engineered QTI import workflows handling packages up to 1 GB with content validation, hash-based duplicate checks and malware scanning before ingestion.
- Reduced authoring time by an estimated 35% with an AI question generator that creates up to 10 variants per template using Angular, .NET, RabbitMQ and Python on AWS Lambda.
- Tuned MySQL procedures with read replicas; fixed memory leaks and accessibility issues in 4 Angular micro-frontends.
- Designed a configurable grading engine and Angular reporting interface for 12 content types, including coordinate geometry questions.
- Integrated Azure Speech pronunciation scoring, fixing memory growth and truncation in long recordings.

PROJECTS

Tenderflow: Tendering & Cost Estimation | ASP.NET Core, Angular, PostgreSQL, Docker

Client project

- Co-developed a scalable tendering platform that replaced spreadsheet-based quoting with automated cost calculations, bulk price imports and price history with rollback.
- Designed the project architecture and built a large part of the Angular frontend, including the main tender-editing screens.
- Implemented role-based access, pre-migration database backups and Jenkins builds for all services.
- Shared ownership of audit logging and a suite of about 490 automated tests.

Payment Ledger: Double-Entry Wallet Backend | .NET 10, PostgreSQL, RabbitMQ

[GitHub](#)

- Developed a double-entry wallet API load-tested at 400 transfers/sec with p95 latency of 26.5 ms and 0 request errors in a 60-second k6 run.
- Secured concurrent transfers with idempotency keys and row locks; 116,018 transactions, 0 unbalanced.
- Introduced a transactional outbox publishing ledger events to RabbitMQ; delivered all 116,018 events and drained a 15,246-event backlog in 39 seconds.

Live Auction: Real-Time Bidding Platform | .NET, SignalR, Redis, PostgreSQL, Angular, Docker

[GitHub](#)

- Developed a horizontally scalable bidding platform delivering live bid updates to 2,000 concurrent users at 81 ms p95 latency with zero errors across 2 server instances.
- Ensured exactly one winner when 20 users placed identical bids at the same instant, with no double-sells and no lost bids.
- Wrote 57 automated tests covering 98% of the business rules.

Docs Q&A: Retrieval-Augmented Question Answering | Python, FastAPI, pgvector, OpenAI

[GitHub](#)

- Built a RAG service streaming cited answers over SSE, fusing vector and full-text search with RRF.
- Added a lease-based ingestion worker that never writes duplicate chunks on crash; tested with pytest and Testcontainers.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, Python, HTML/CSS

Backend: .NET 8/10, ASP.NET Core, FastAPI, REST APIs, SignalR, RabbitMQ, Redis, EF Core, Dapper

Frontend: Angular, RxJS, Angular Material, micro-frontends, WCAG accessibility

Databases: MySQL, PostgreSQL, pgvector, MongoDB, Elasticsearch

Cloud & DevOps: AWS (Lambda, EC2, S3, RDS, CloudWatch), Docker, GitHub Actions, CI/CD, Git, OpenTelemetry

Testing & AI: xUnit, pytest, Testcontainers, Vitest, RAG, OpenAI API, Azure Speech SDK

EDUCATION

Government Engineering College, Gandhinagar

B.E. in Computer Engineering, CGPA 8.15/10

Gandhinagar, India

2020 – 2024